Quantum Teleportation:
Protocol, Implementation, and Applications

Ying-Han Su, Ting-Peng Huang, Po-Rui Chen[1,*]

[1] *Department of Physics, National Taiwan University, Taipei, Taiwan 10617, R.O.C.*

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Abstract

This report provides a comprehensive study of quantum teleportation, covering its protocol, implementation, and applications. We begin with an introduction to the fundamental concepts and historical development of quantum teleportation. We then present the theoretical framework and the step-by-step protocol for achieving quantum teleportation. Following this, we discuss various implementation strategies and their practical challenges. Finally, we explore the current and potential applications of quantum teleportation in quantum information processing and communication.

* e-mail: b11303052@ntu.edu.tw

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I. INTRODUCTION

Quantum teleportation is one of the most striking protocols in quantum information science. Despite its name, it does not mean transporting matter from one place to another. Instead, it refers to the transfer of an unknown quantum state from one physical system to another, without directly sending the original system itself. This is surprising because an unknown quantum state cannot be fully determined by a single measurement, and it also cannot be copied perfectly due to the no-cloning theorem [1, 2]. Therefore, quantum teleportation addresses a fundamental question: how can an unknown quantum state be transferred if it cannot be measured completely or cloned?

The theoretical protocol was first proposed by Bennett *et al.* in 1993 [3]. Their result showed that an unknown qubit can be transferred from Alice to Bob using two resources: a shared entangled pair and two bits of classical communication. Alice performs a joint Bell-state measurement on the unknown qubit and her half of the entangled pair, then sends the classical result to Bob. Depending on this result, Bob applies the appropriate Pauli correction to recover the original state. The protocol therefore demonstrates that entanglement can be used as a physical resource for quantum information processing.

The first experimental demonstrations soon followed. Bouwmeester *et al.* realized quantum teleportation using photonic qubits and Bell-state measurement [4], while Boschi *et al.* implemented teleportation of an unknown pure state using a dual-degree-of-freedom optical setup [5]. Since then, teleportation has developed from a foundational protocol into an important tool for quantum communication. Long-distance demonstrations include active feed-forward teleportation over 143 km in free space [6], and ground-to-satellite teleportation over distances up to 1400 km [7]. These experiments are closely connected to the long-term goal of building a quantum internet, where quantum states and entanglement can be distributed between distant nodes [8]. Quantum repeaters, which combine entanglement distribution, purification, and teleportation, are one possible route toward overcoming loss in long-distance quantum communication [9].

In this report, we first introduce the necessary mathematical background, including the Bell state and Bell measurement, to derive the protocol identity. Then, we focus on the optical experiment proposed by Boschi *et al.* Finally, we discuss the applications of teleportation in quantum communication, quantum networks, and quantum computing.

II. MATHEMATICAL BACKGROUND

A. Qubits and the Hilbert Space Formalism

The fundamental unit of quantum information is called a qubit. Unlike a classical bit, which takes the value of either 0 or 1 at a given time, a qubit exists in a two-dimensional Hilbert space $\mathcal{H} = \text{span}\{|0\rangle, |1\rangle\} \cong \mathbb{C}^2$, and its state can be described by a unit vector with an orthonormal computational basis $\{|0\rangle, |1\rangle\}$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \ \alpha, \beta \in \mathbb{C}, \ |\alpha|^2 + |\beta|^2 = 1 \tag{1}$$

where α and β are the probability amplitudes: measurement yielding $|0\rangle$ or $|1\rangle$ has a probability $|\alpha|^2$ and $|\beta|^2$, respectively [10]. Unlike a classical bit, a qubit can exist in a superposition of both states, which has no classical analogue. Physical realizations of qubits include the two polarization states of a photon and two energy levels of an atom or ion.

Bloch Sphere Since a global phase factor $e^{i\gamma}$ does not have observable effect, Eq. 1 can be written in a more general expression

$$|\psi\rangle = \cos\frac{\theta}{2}\,|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}\,|1\rangle\,, \ \theta \in [0, \pi], \ \varphi \in [0, 2\pi) \tag{2}$$

The two parameters θ and φ define a unique point on the unit three-dimensional sphere, known as the Bloch sphere as shown in Figure 1. The north pole ($\theta = 0$) corresponds to $|0\rangle$, while the south pole corresponds to $|1\rangle$, and every point on the equator represents an equal superposition such as $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.

Multi-qubit systems. For a system of two qubits, we use the tensor product of the individual Hilbert space $\mathcal{H}_{12} = \mathcal{H}_1 \otimes \mathcal{H}_2 \cong \mathbb{C}^4$ to represent the state space, with computational basis kets $\{|00\rangle\,, |01\rangle\,, |10\rangle\,, |11\rangle\}$. Here we use the ordering convention. That is, $|01\rangle = |0\rangle_1 \otimes |1\rangle_2$. Similarly, if it is a system of three qubits, then $|010\rangle = |0\rangle_1 \otimes |1\rangle_2 \otimes |0\rangle_3$. A general two-qubit state takes the form

$$|\psi\rangle = \alpha_{00}\,|00\rangle + \alpha_{01}\,|01\rangle + \alpha_{10}\,|10\rangle + \alpha_{11}\,|11\rangle\,, \ \sum_x |\alpha_x|^2 = 1 \tag{3}$$

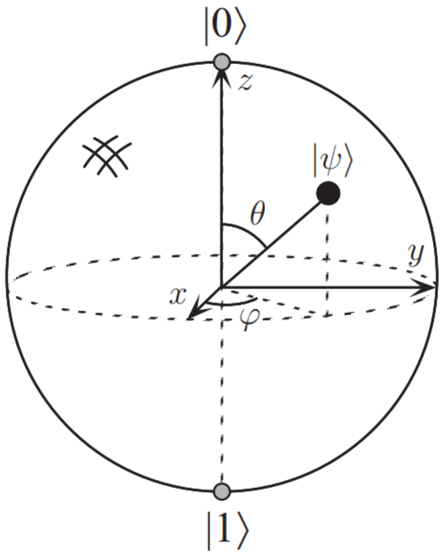

FIG. 1: Bloch sphere schematic. An arbitrary qubit can be represented by one point on the sphere [10].

B. Entanglement and the Bell States

A two-qubit state is called separable if it can be factored as $|\psi\phi\rangle = |\psi\rangle_A \otimes |\phi\rangle_B$. This means that qubit A is in state $|\psi\rangle$, while qubit B is in state $|\phi\rangle$. On the other hand, an entangled state cannot be factorized into the product of two single-qubit states. In this case, the quantum state can only be understood as a property of the whole system. The most important entangled states for quantum teleportation are the Bell states [3, 10]

$$\begin{aligned} |\Phi^+\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \qquad |\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \\ |\Psi^+\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \qquad |\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle). \end{aligned} \tag{4}$$

These four states form an orthonormal basis for the two-qubit Hilbert space. These states are also maximally entangled as tracing out either qubit leaves the reduced density matrix $\rho = I/2$, where I is the identity matrix. For example, consider the Bell state $|\Phi^+\rangle_{AB}$, then the corresponding density matrix is $\rho_{AB} = |\Phi^+\rangle\langle\Phi^+|$. Now, if we only look at the qubit A subsystem, we trace out qubit B to get the reduced density matrix as

$$\rho_A = \mathrm{tr}_B(\rho_{AB}) = \frac{I_A}{2}. \tag{5}$$

Similarly, we have $\rho_B = \frac{I_B}{2}$. This implies that when measuring either qubit, we get equal probability of getting $|0\rangle$ or $|1\rangle$. However, the measurement results of system A and system B are perfectly correlated. For example, if measuring qubit A gives $|0\rangle$, then measuring qubit B in the same basis gives $|0\rangle$. Similarly, if A gives $|1\rangle$, then B also gives $|1\rangle$. That is, each local measurement result is random, but it becomes highly correlated once we look at the system as a whole.

C. Quantum Measurement and Bell Measurement

The projection postulate states that measurement irreversibly collapses the superposition state into one of its components [10]. Therefore, measurement on the qubit is a physical process that will disturb the original state. A projective measurement is described by a Hermitian observable $M = \sum_m m P_m$, where $\{P_m\}$ are orthogonal projection operators onto the eigenspace of M with eigenvalue m and satisfy $\sum_m P_m = I$ and $P_m P_{m'} = \delta_{mm'} P_m$. Given a pure state $|\psi\rangle$, the probability of obtaining outcome m is

$$p(m) = \langle\psi| P_m |\psi\rangle \tag{6}$$

and the post-measurement state collapses to [10]

$$|\psi_m\rangle = \frac{P_m |\psi\rangle}{\sqrt{p(m)}}. \tag{7}$$

For example, consider a qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. If we do the measurement in the computational basis $\{|0\rangle, |1\rangle\}$, then the corresponding projection operators are $P_0 = |0\rangle\langle 0|$, $P_1 = |1\rangle\langle 1|$. The probability and the post-measurement states of the two possible outcomes are

$$\begin{aligned} p(0) &= \langle\psi| P_0 |\psi\rangle = |\alpha|^2, \qquad P_0|\psi\rangle = |0\rangle\langle 0|(\alpha|0\rangle + \beta|1\rangle) = \alpha|0\rangle \\ p(1) &= \langle\psi| P_1 |\psi\rangle = |\beta|^2, \qquad P_1|\psi\rangle = |1\rangle\langle 1|(\alpha|0\rangle + \beta|1\rangle) = \beta|1\rangle. \end{aligned} \tag{8}$$

The projection P_0 and P_1 keep only $|0\rangle$ and $|1\rangle$ components, respectively. In other words, the post-measurement state becomes either $|0\rangle$ or $|1\rangle$. Therefore, the original superposition is generally destroyed. For the teleportation protocol, we also need a two-qubit projective measurement called the Bell measurement. Instead of measuring each qubit separately in the computational basis, a Bell measurement projects the two-qubit system onto the Bell basis. Its role in teleportation will be discussed in the next section. Similar to the projective measurement, we define the Bell-measurement projectors as

$$P_k = |\beta_k\rangle \langle\beta_k| , \ |\beta_k\rangle \in \{|\Psi^{\pm}\rangle , |\Phi^{\pm}\rangle\}. \tag{9}$$

Thus, a Bell measurement yields one of the four Bell-state outcomes. That is, computational-basis measurement tells us whether each qubit is individually in $|0\rangle$ or $|1\rangle$, while a Bell measurement determines which Bell state the two-qubit system occupies as a whole.

D. Pauli Operators and Unitary Corrections

The Pauli operators are a set of 2×2 unitary matrices forming a basis for the space of single-qubit operators [10]:

$$\begin{aligned} I &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad X = \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \\ Z = \sigma_3 &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad Y = \sigma_2 = iXZ = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \end{aligned} \tag{10}$$

These are the standard single-qubit Pauli operators used in quantum information. In particular, X is a bit-flip operator ($X|0\rangle = |1\rangle , \ X|1\rangle = |0\rangle$), Z acts as a phase-flip operator ($Z|0\rangle = |0\rangle , \ Z|1\rangle = -|1\rangle$), while Y is the same operator as XZ up to a phase factor. Therefore, in the teleportation protocol below, we will use XZ as the fourth correction operation. These four operators $\{I, \ X, \ Z, \ XZ\}$ will serve as the unitary correction operators for Bob to recover the original quantum state after Alice's Bell measurement.

III. STANDARD QUANTUM TELEPORTATION PROTOCOL

In this section, we use the mathematical tools introduced above to derive the standard quantum teleportation protocol. The goal of the protocol is to transfer an unknown quantum state from Alice to Bob without physically sending the original qubit through a quantum channel. Instead, the protocol uses one shared Bell pair, one Bell measurement, two bits of classical communication, and a conditional Pauli correction.

Suppose Alice has an unknown qubit state

$$|\psi\rangle_1 = \alpha\,|0\rangle_1 + \beta\,|1\rangle_1\,,\;\; |\alpha|^2 + |\beta|^2 = 1. \tag{11}$$

Here the subscript 1 labels the unknown input qubit. Alice does not know the coefficients α and β, and she cannot determine them by a direct measurement because the state would collapse. Before the protocol begins, Alice and Bob are assumed to share an entangled Bell pair, say

$$|\Phi^+\rangle_{23} = \frac{1}{\sqrt{2}}(|00\rangle_{23} + |11\rangle_{23}), \tag{12}$$

where qubits 2 and 3 are held by Alice and Bob, respectively. The total initial state of the three-qubit system is

$$\begin{aligned} |\Psi\rangle_{123} &= |\psi\rangle_1\,|\Phi^+\rangle_{23} \\ &= \frac{1}{\sqrt{2}}\left(\alpha\,|000\rangle_{123} + \alpha\,|011\rangle_{123} + \beta\,|100\rangle_{123} + \beta\,|111\rangle_{123}\right), \end{aligned} \tag{13}$$

where in the second line of the equation we expand the total qubit state. Since Alice will perform a Bell measurement on qubits 1 and 2, it is useful to rewrite the state in the Bell basis of these two qubits as

$$\begin{aligned} |00\rangle_{12} &= \frac{1}{\sqrt{2}}\left(|\Phi^+\rangle_{12} + |\Phi^-\rangle_{12}\right), \qquad & |11\rangle_{12} &= \frac{1}{\sqrt{2}}\left(|\Phi^+\rangle_{12} - |\Phi^-\rangle_{12}\right), \\ |01\rangle_{12} &= \frac{1}{\sqrt{2}}\left(|\Psi^+\rangle_{12} + |\Psi^-\rangle_{12}\right), \qquad & |10\rangle_{12} &= \frac{1}{\sqrt{2}}\left(|\Psi^+\rangle_{12} - |\Psi^-\rangle_{12}\right). \end{aligned} \tag{14}$$

Thus, the total state becomes

$$
\begin{aligned}
|\psi\rangle_1 |\Phi^+\rangle_{23} = \frac{1}{2}\Big[& |\Phi^+\rangle_{12} (\alpha |0\rangle_3 + \beta |1\rangle_3) + |\Phi^-\rangle_{12} (\alpha |0\rangle_3 - \beta |1\rangle_3) \\
& + |\Psi^+\rangle_{12} (\beta |0\rangle_3 + \alpha |1\rangle_3) + |\Psi^-\rangle_{12} (-\beta |0\rangle_3 + \alpha |1\rangle_3) \Big].
\end{aligned} \tag{15}
$$

We may further rewrite this expression using the Pauli operators as

$$
|\psi\rangle_1 |\Phi^+\rangle_{23} = \frac{1}{2}\left[|\Phi^+\rangle_{12} |\psi\rangle_3 + |\Phi^-\rangle_{12} Z |\psi\rangle_3 + |\Psi^+\rangle_{12} X |\psi\rangle_3 + |\Psi^-\rangle_{12} XZ |\psi\rangle_3 \right]. \tag{16}
$$

This equation is the central identity of quantum teleportation. It shows that when qubits 1 and 2 are written in the Bell basis, Bob's qubit 3 is related to the original input state by one of four possible Pauli operations. Alice now performs a Bell measurement on qubits 1 and 2. According to the projection postulate, the total state collapses into one of the four Bell states with equal probability. If Alice obtains the outcome $|\Phi^+\rangle_{12}$, then Bob's qubit is already $|\psi\rangle_3$. After Alice communicates this outcome, Bob knows that the correct operation is simply I. If Alice gets $|\Phi^-\rangle_{12}$, then Bob's qubit is $Z |\psi\rangle_3$. Bob needs to apply Z so that $Z(Z |\psi\rangle_3) = Z^2 |\psi\rangle_3 = |\psi\rangle_3$. Similarly, if Alice obtains $|\Psi^+\rangle_{12}$, Bob gets $X |\psi\rangle_3$ and needs to apply X so that $X(X |\psi\rangle_3) = X^2 |\psi\rangle_3 = |\psi\rangle_3$. Finally, if Alice obtains $|\Psi^-\rangle_{12}$, Bob applies ZX such that $ZX(XZ |\psi\rangle_3) = |\psi\rangle_3$. These four cases can be summarized into the following table as

Alice's Bell outcome	Bob's state before correction	Bob's correction
$\lvert\Phi^+\rangle_{12}$	$\lvert\psi\rangle_3$	I
$\lvert\Phi^-\rangle_{12}$	$Z \lvert\psi\rangle_3$	Z
$\lvert\Psi^+\rangle_{12}$	$X \lvert\psi\rangle_3$	X
$\lvert\Psi^-\rangle_{12}$	$XZ \lvert\psi\rangle_3$	ZX

(17)

After Alice's measurement, she needs to tell Bob which one of these four Bell states she got. Since one classical bit has two possible values, 0 and 1, two classical bits can represent four possible messages: 00, 01, 10, and 11. This is enough to label the four Bell-measurement outcomes. For example, they may agree that 00 corresponds to $|\Phi^+\rangle_{12}$. Alice sends this two-bit message to Bob through a classical channel. Once Bob receives it, he knows which Pauli correction to apply and can recover the original unknown state. In resource language, the teleportation protocol can be summarized as

$$1 \text{ ebit} + 2 \text{ classical bits} \to 1 \text{ transmitted qubit}. \tag{18}$$

IV. PHYSICAL IMPLICATIONS OF QUANTUM TELEPORTATION

After deriving the standard teleportation protocol, it is important to clarify what the protocol does and does not imply physically. Quantum teleportation transfers an unknown quantum state from Alice's qubit to Bob's qubit, but it does not transfer matter, does not create a second copy of the state, and does not allow faster-than-light communication. The protocol works only by combining a shared entangled pair with a two-bit classical communication channel. In this section, we discuss these physical implications and explain why the protocol is consistent with the basic principles of quantum mechanics and relativity.

A. No-Cloning

Quantum teleportation does not create a second copy of the unknown quantum state. Although Bob's final qubit becomes $|\psi\rangle_3 = \alpha\,|0\rangle_3 + \beta\,|1\rangle_3$, Alice's original qubit is no longer available after the Bell measurement. The Bell measurement projects qubits 1 and 2 into one of the four Bell states, and therefore irreversibly destroys the original unknown state on Alice's side. This is why quantum teleportation is consistent with the no-cloning theorem.

The no-cloning theorem, first formulated by Wootters and Zurek [1] and independently by Dieks [2], states that it is impossible to make an exact copy of an arbitrary unknown quantum state. The proof is straightforward. Suppose there exists a universal unitary cloning machine U such that $U\,|\psi\rangle\,|s\rangle = |\psi\rangle\,|\psi\rangle$ for any unknown state $|\psi\rangle$, where $|s\rangle$ is a blank initial state. If the same machine can also clone another state $|\phi\rangle$, then $U\,|\phi\rangle\,|s\rangle = |\phi\rangle\,|\phi\rangle$. Taking the inner product of these two equations gives $\langle\psi|\phi\rangle = \langle\psi|\phi\rangle^2$. This condition is only satisfied when $\langle\psi|\phi\rangle = 0$ or $\langle\psi|\phi\rangle = 1$. In other words, the two states must be either orthogonal or identical. Therefore, no single unitary operation can clone an arbitrary unknown quantum state [10].

Teleportation avoids this contradiction because it does not copy the state. Instead, the quantum information initially contained in qubit 1 is transferred to qubit 3, while the original state at Alice's side is destroyed by measurement. Therefore, teleportation should be understood as state transfer rather than state duplication.

B. No Faster-Than-Light Communication

Quantum teleportation also does not allow faster-than-light communication. Although the shared Bell pair provides nonclassical correlations between Alice and Bob, Bob cannot recover the unknown state before receiving Alice's classical message. That is, Bob needs Alice's message so that he knows what operation he should apply to recover the unknown qubit. This can be made more precise by looking at Bob's reduced density matrix before the classical message arrives. Since the four Bell-measurement outcomes occur with equal probability, Bob's local state is an average over the four possible states:

$$\rho_3 = \frac{1}{4}\left(|\psi\rangle\langle\psi| + Z|\psi\rangle\langle\psi|Z + X|\psi\rangle\langle\psi|X + XZ|\psi\rangle\langle\psi|ZX\right). \tag{19}$$

For an arbitrary qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, this average becomes

$$\rho_3 = \frac{I}{2}. \tag{20}$$

Thus, before receiving Alice's classical message, Bob's local state is completely mixed and independent of α and β. Any measurement Bob performs locally gives random results and reveals no information about the unknown input state. Bennett *et al.* [3] emphasized this point in the original teleportation paper: if Bob tries to complete the protocol by guessing Alice's classical message before it arrives, his state is only a maximally mixed state and gives no information about the input state. Therefore, Bob cannot extract any information about $|\psi\rangle$ before receiving Alice's two-bit message. Since that message must be sent through a classical channel, teleportation cannot be used for faster-than-light communication.

C. Security of the Classical Message

Alice's two classical bits are necessary for Bob to complete the teleportation protocol, but they do not carry the unknown quantum state itself. They only specify which Bell-measurement outcome Alice obtained, and therefore which Pauli correction Bob should apply.

This follows from Eq. 16. Alice's Bell measurement has four possible outcomes, each occurring with probability 1/4, independent of the input state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. Thus, the classical message is statistically independent of α and β. If an eavesdropper intercepts only the classical channel, they may know whether Alice sent 00, 01, 10, or 11, but this only reveals the correction label, not the teleported state.

The shared Bell pair alone also does not contain the unknown state, since it is prepared before Alice receives $|\psi\rangle$. Bob can reconstruct the state only when the shared entanglement, Alice's Bell measurement, the two-bit classical message, and the corresponding Pauli correction are combined.

This separation between classical communication and quantum information is one reason why teleportation is a central primitive in quantum communication and quantum networks [11]. This does not mean that teleportation by itself is a complete encryption protocol. Rather, it shows that the classical message alone carries no information about the unknown quantum amplitudes. The reconstruction of the state requires the full protocol, including the shared entanglement.

V. EXPERIMENTAL IMPLEMENTATION VIA DUAL-PHOTON ENTANGLEMENT

A. Experimental Background

The 1998 experiment by Boschi *et al.* provided an alternative approach for realizing quantum information protocols. While the original Bennett *et al.* (1993) model required three independent photons—which introduced technical challenges regarding temporal synchronization—this setup utilized a two-photon, dual-degree-of-freedom architecture. By utilizing both the momentum (path or k-vector) and polarization of a single entangled photon pair, researchers accessed a four-dimensional Hilbert space (2 paths$\times$2 polarizations) within a bipartite system. This configuration addressed two primary limitations of prior multi-photon experiments:

- **Complete Bell State Measurement:** Unlike linear-optics three-photon schemes (which are bounded by a 50% success rate), this approach permits projection onto the complete Bell basis, enabling the identification of all four states simultaneously.
- **Reduction of Coincidence Requirements:** Deriving the process from a single entangled photon pair (EPR source) avoids the difficulty of synchronizing independent sources within sub-nanosecond coincidence windows.

The experimental execution begins at the EPR source, where the quantum channel is established.

B. Implementation of the Dual-Degree-of-Freedom Setup

1. *Generation of the EPR Entangled Source*

Entanglement is generated via Type-II spontaneous parametric down-conversion (SPDC). Specifically, a 200 mW, 351.1 nm ultraviolet argon laser pumps a β-barium borate (BBO) crystal, producing polarization-entangled photon pairs at 702.2 nm. To facilitate the dual-degree architecture, this initial polarization entanglement is converted into spatial path entanglement via birefringent calcite crystals (C), which split the incident photons into discrete spatial paths (a and b). The resulting EPR state is described by:

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|a_1\rangle|a_2\rangle + |b_1\rangle|b_2\rangle)$$

Photon 1 is routed toward Alice's station, while photon 2 traverses backreflected paths a_2 and b_2 through the BBO crystal toward Bob.

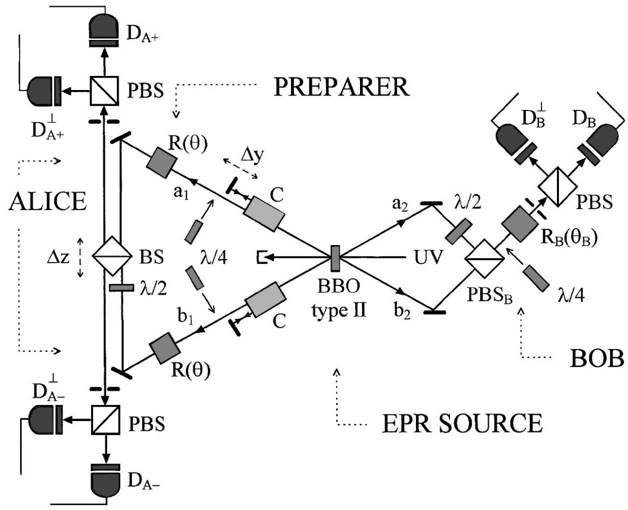

FIG. 2: Diagram of experimental setup showing the separate roles of the preparer, Alice, and Bob.

2. *Preparation and Encoding of the Unknown State*

A logical distinction is maintained between the Preparer and Alice. To ensure the state $|\phi\rangle$ remains unknown to Alice, the Preparer encodes it onto photon 1's polarization before it reaches Alice's apparatus. This relies on local unitary operations on paths a_1 and b_1:

- **Polarization Initialization:** Photon 1 enters with an initial defined polarization (typically vertical, $|v\rangle_1$).
- **Unitary Manipulation:** Quarter-wave plates and Fresnel rhomb rotators (R) transform this into a general pure state: $|\phi\rangle = \alpha|v\rangle_1 + \beta|h\rangle_1$.
- **Path Independence:** Operating identically on paths a_1 and b_1, the polarization state effectively factors out of the path entanglement.

This yields a global tripartite system where the unknown polarization is carried by the path-entangled photon 1:

$$|\Psi_{total}\rangle = \frac{1}{\sqrt{2}}(|a_1\rangle|a_2\rangle + |b_1\rangle|b_2\rangle)(\alpha|v\rangle_1 + \beta|h\rangle_1)|h\rangle_2$$

*3. **Interferometric Bell State Measurement***

Alices Bell State Measurement (BSM) interferometer projects photon 1's bipartite state onto four orthogonal path-polarization Bell states:

$$|c^{\pm}\rangle = \frac{1}{\sqrt{2}}(|a_1\rangle|v\rangle_1 \pm |b_1\rangle|h\rangle_1)$$

$$|d^{\pm}\rangle = \frac{1}{\sqrt{2}}(|a_1\rangle|h\rangle_1 \pm |b_1\rangle|v\rangle_1)$$

Prior to entering a 50:50 beam splitter (BS), a Fresnel rhomb in path b_1 induces a 90° polarization rotation. This ensures path and polarization degrees of freedom are properly aligned for interference. The BS then acts as a unitary operator, mapping distinct path-polarization correlations to specific output ports.

The BS is mounted on a micrometrical stage to precisely tune the path-length difference (Δz). Alignment is confirmed by interference dips in coincidence count rates, signifying spatiotemporal wave packet overlap.

Mechanics of Detection: Polarizing beam splitters (PBS) sort the emerging photons into four avalanche photodiode detectors ($D_{A+}, D^{\perp}_{A+}, D_{A-}, D^{\perp}_{A-}$). Detector coincidences project onto the Bell basis:

- A detection at $D^{\perp}_{A+}$ or $D^{\perp}_{A-}$ identifies states $|c^{\pm}\rangle$.
- A detection at D_{A+} or D_{A-} identifies states $|d^{\pm}\rangle$.

This measurement determines the unitary operation Bob must perform to reconstruct the initial state.

4. Passive Reconstruction and State Verification

At the receiving station, photon 2 arrives via trajectories a_2 and b_2. To resolve the path entanglement, polarization in path a_2 is orthogonally rotated by 90° via a half-wave plate before recombining with path b_2 at the polarizing beam splitter (PBS_B).

Bob employs *passive reconstruction.* Rather than dynamically rotating the state using active electro-optic switches, he reorients his measurement basis to verify the teleported state based on Alice's classical transmission.

Unitary Transformation Logic:

- **Linear States:** Bob adjusts his Fresnel rhomb (R_B) to an angle θ_B, determined by Alices Bell state outcome.
- **Elliptical States:** Bob inserts a quarter-wave plate at angle γ_B to map the expected elliptical state back into an observable linear basis.
- **Phase Matching:** Bob uses a micrometrical stage to tune Δy, ensuring paths satisfy the phase-matching criteria for high-fidelity verification.

C. Verification and Comparison with the Classical Limit

To verify quantum teleportation, the experimental fidelity (S)—representing the average probability of the teleported state passing verification—must exceed the classical communication limit of 0.75. Exceeding this threshold indicates that entanglement was utilized as a non-local resource.

In the 1998 experiment, S was derived through a sampling process across three equally spaced linear polarization settings (0°, 120°, and −120°). To ensure a representative result on the Bloch sphere, each of these three states was assigned a weight of 1/3, while each of the four possible Bell state measurement outcomes was weighted equally at 1/4. The final observed fidelity of 0.853 ± 0.012 exceeds the classical bound by eight standard deviations. This result confirms that the dual-degree-of-freedom architecture functioned effectively, consistent with the successful teleportation of unknown pure quantum states.

VI. VARIANTS OF QUANTUM TELEPORTATION

A. Optical Modes System: Continuous-Variable Teleportation

B. Atomics Ensembles and Trapped Ion System

C. Solid-State System

VII. ADVANCES IN QUANTUM TELEPORTATION

A. Quantum Teleportation for Quantum Communication

B. Quantum Gate Teleportation for Quantum Computing

VIII. OUTLOOK OF QUANTUM TELEPORTATION

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